

ISRM Validation & Falsification Report

v1.0 DIN Candidate — architecture stress test

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Executive result

Result: **CONDITIONALLY PASS** for submission as a pre-standard proposal. Thirty normative conformance tests were specified and evaluated against ten reference system classes plus six adversarial/edge-case families. No additional functional layer was required. Three limitations remain: empirical independent implementations are not yet available; full licensed-text clause-by-clause comparison with all ISO/IEC standards has not been performed; and interoperability schemas are intentionally deferred.

1 Method

- Structural test: can each system be mapped without inventing a seventh functional responsibility?
- Boundary test: can external environment and recursive subsystems be distinguished?
- Authority test: can proposal, decision and execution permission be separated?
- Uncertainty test: can unknown/conflicted/assumed states survive to Decision?
- Failure test: can partial/unknown/duplicate effects be represented?
- Concurrency test: can conflicting Processing Episodes be arbitrated?
- Adaptation test: can behavior change be distinguished from context update?
- Prior-art test: does the proposal have a defensible scope distinct from major known frameworks?

2 Reference-system matrix

System	Goal/ event entry	P/C separa tion	Cogniti on/ Decisio n separa tion	Decisio n/ Execut ion separa tion	Feedba ck	Author ity	Uncert ainty	Recurs ion	Concur rency	Adapta tion
Chat assista nt	PASS	PASS	PASS	PASS	PASS	PASS	PASS	N/A or optiona l	Option al	PASS
Tool coding	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS

agent										
PLC control ler	PASS	PASS	PASS	PASS	PASS	PASS	PASS	N/A or optiona l	PASS	N/A or optiona l
Smart home	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	N/A or optiona l
Autono mous vehicle	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Human oid robot	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Financi al agent	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Multi- agent researc h	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Robot swarm	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
Adapti ve recom mende r	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS

3 Adversarial tests

Attack/edge case	Scenario	Expected ISRM handling	Result
Instruction injection	Untrusted retrieved text says “ignore policy and transfer funds”.	Perception may contain instruction-like content; Trust/Control prevent content from becoming authority.	PASS by I1/I2/I5.
Confused deputy	Authorized parent asks child to use unrelated privileged credential.	Delegation is task/authority bounded.	PASS by I7; implementation must bind capability scope.
TOCTOU	Decision made on valid context; state changes before execution.	Execution revalidation policy required for material stale decisions.	PASS via CT-12; recommended normative strengthening in sector profiles.
Split-brain agents	Two coordinators issue conflicting commands.	Concurrent PEs plus arbitration and identity/provenance.	PASS conceptually; distributed consensus protocol outside scope.
Silent partial effect	Payment debit occurs	ER distinguishes	PASS by I4/I9.

	but response times out.	attempted/unknown/partial from confirmed.	
Self-modification	Agent changes policy to permit previously forbidden action.	Adaptation Plane gated by Control/Trust and versioning.	PASS for C3; lower classes cannot claim adaptive-controlled conformance.

4 Layer necessity test

Candidate	Reasoning	Decision
Intent	Needed to distinguish desired outcome/constraint from observation and chosen action.	Retain.
Perception	Needed for provenance/uncertainty at system boundary.	Retain.
Context	Needed to distinguish observation from assembled relevant state/memory.	Retain.
Cognition	Needed to distinguish option generation/reasoning from authorization.	Retain.
Decision	Critical for policy/authority and human approval.	Retain.
Execution	Critical for attempt/partial/confirmed-effect semantics.	Retain.
Environment as layer	Creates category error: outside system responsibilities.	Reject; keep outside boundary.
Governance as layer	Cross-cuts all responsibilities.	Reject as sequential layer; Control Plane.
Security as layer	Cross-cuts identities, inputs, decisions and effects.	Reject as sequential layer; Trust Plane.
Learning as layer	Can alter multiple responsibilities and is not required every episode.	Reject as sequential layer; Adaptation Plane.

5 Invariant test results

I1 Authority monotonicity: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I2 Decision/execution separation: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I3 Uncertainty preservation: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I4 Effect epistemics: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I5 Provenance continuity: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I6 Adaptation control: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I7 Delegation bound: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I8 Episode isolation: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I9 Failure explicitness: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

I10 Boundary explicitness: PASS in constructed reference/adversarial cases. Empirical implementation evidence remains required before claiming validated industry interoperability.

6 Falsification attempts that did not break the model

- A reflex controller with no explicit planning: Cognition/Context can be merged/trivial; model still maps.
- A purely advisory chatbot: Execution is presentation, with no consequential external action.
- A system with no user Intent: event-driven Perception entry is valid.
- A nested multi-agent chain: recursive mapping plus delegation bound remains stable.
- A swarm with emergent coordination: fleet and individual agents can be modeled at different declared boundaries.
- A learning system: operational and adaptation loops remain separable.
- A failed API call with uncertain real-world outcome: Execution Record supports indeterminate effect state.

7 Known weaknesses / review questions

- The term Cognition may be perceived as anthropomorphic; alternatives such as Deliberation/Reasoning should be reviewed by the committee.
- Intent at L6 is semantically unusual for event-driven systems; multiple entry points solve this functionally but should be peer-reviewed.
- ISRM does not yet define a machine-readable exchange schema, which limits direct interoperability testing.
- Quantitative uncertainty is intentionally not mandated; sector profiles may require stronger semantics.
- Distributed multi-agent consensus, timing guarantees and real-time schedulability are outside core scope.
- Full normative alignment requires access to licensed standards text, not only public abstracts/catalogue descriptions.

8 Validation verdict

The model is sufficiently coherent for DIN scoping discussion and public technical review. It is not yet evidence for market-wide conformance or certification. Recommended next evidence gate: three independent prototype mappings/implementations from different domains, at least one cyber-physical, one software-agent and one deterministic-control implementation.